INCH-POUND

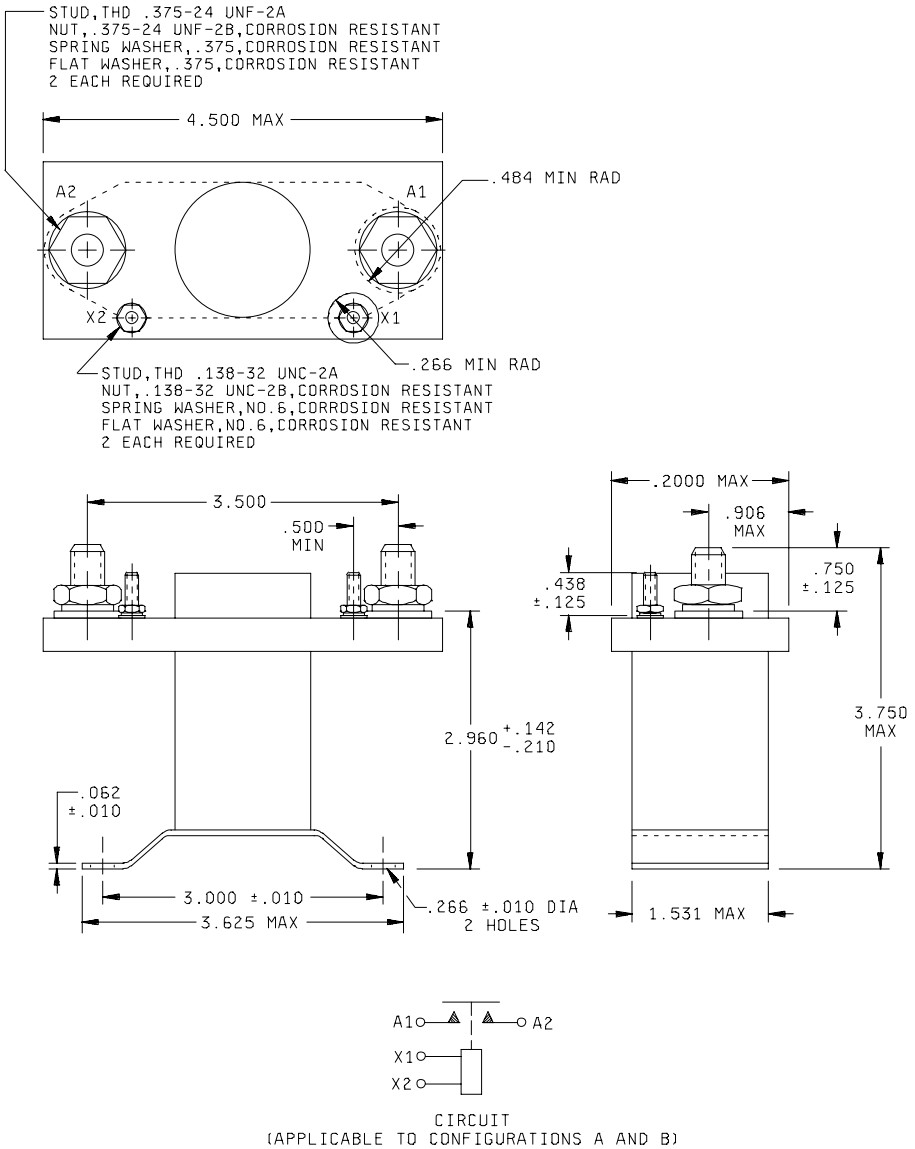
MS24172J
15 April 2003
SUPERSEDING
MS24172H
25 July 1990

DETAIL SPECIFICATION SHEET

# RELAYS, ELECTROMAGNETIC, 200 AMPERES, 1 PST (N.O.), TYPE III, NONHERMETICALLY SEALED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the relay described herein shall consist of this specification and the latest issue of MIL-PRF-6106.

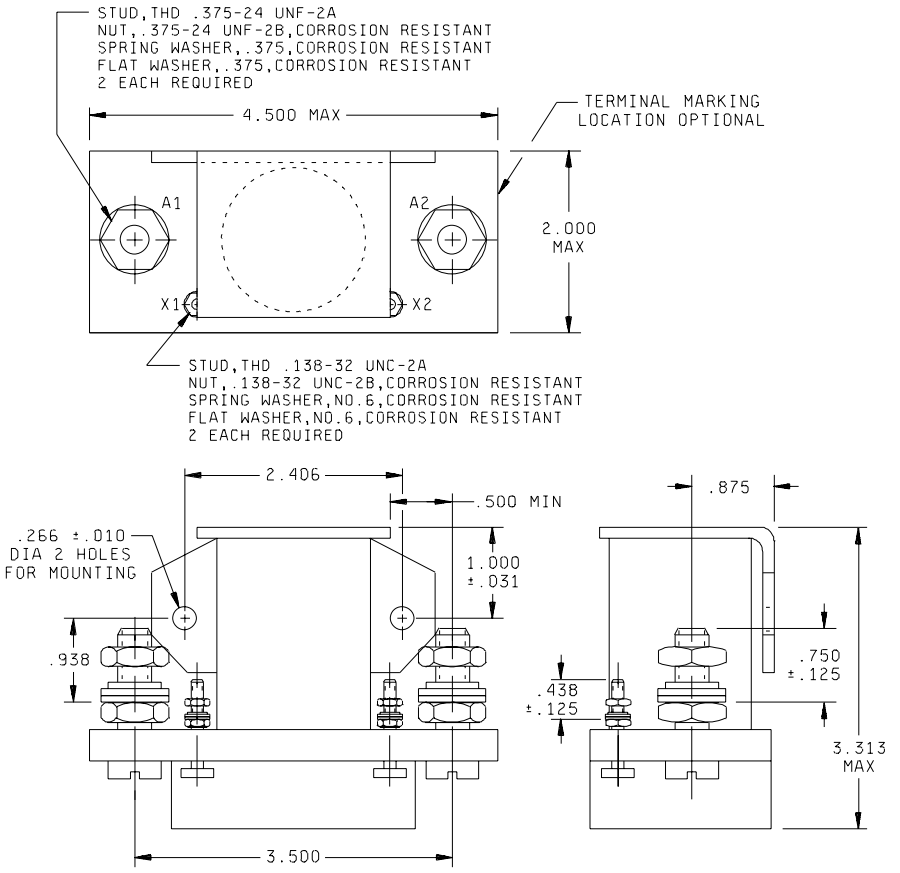



CIRCUIT
(APPLICABLE TO CONFIGURATIONS A AND B)

CONFIGURATION A. Outline drawing (for details see tables I and II).


DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

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CONFIGURATION B. Outline drawing (for details see tables I and II).

| Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|
| .010 | 0.25 | .375 | 9.52 | .906 | 23.01 | 3.000 | 76.20 |
| .062 | 1.57 | .438 | 11.13 | .938 | 23.83 | 3.313 | 84.15 |
| .125 | 3.18 | .484 | 12.30 | 1.000 | 25.40 | 3.500 | 88.90 |
| .138 | 3.51 | .500 | 12.70 | 1.531 | 38.89 | 3.750 | 95.26 |
| .142 | 3.61 | .750 | 19.06 | 2.000 | 50.80 | 4.500 | 114.30 |
| .210 | 5.33 | .813 | 20.66 | 2.406 | 61.11 | | |
| .266 | 6.76 | .875 | 22.22 | 2.960 | 75.18 | | |

NOTES:
1. Dimensions are in inches.
2. Terminal cover not required.
3. Additional flat washer may be used for terminal seat.
4. Part number MS24172-D1 replaces part no. MS24172-1.
5. This specification sheet takes precedence over documents referenced herein.
6. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.
7. Metric equivalents are given for general information only.
8. Unless otherwise specified, tolerances are ±.062 (1.57 mm).
9. Terminal temperature rise under continuous current conditions, 95°C. Intermediate current to be conducted at 71°C.
10. Shape of relay optional within envelope dimensions.
11. Cadmium or cadmium compounds are prohibited on external hardware. A transition period to non-cadmium hardware is authorized for up to 1 year from the date of this revision.
12. Spring washer on drawing is a spring lock washer.



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TABLE I. Dash numbers and characteristics.

| Dash number MS24172- | Type | Coil | Terminal type | Mounting or mating socket | Max weight in pounds |
|---|---|---|---|---|---|
| D1 1/ | III | dc | Stud | Bracket - bottom | 1.33 |
| D2 2/ | III | dc | Stud | Bracket - side | 1.33 |

1/ For Government logistics support MS24172-D1 shall be used in lieu of AN3371-2.
2/ For Government logistics support MS24172-D2 shall be used in lieu of AN3371-1.

TABLE II. Operating characteristics.

| PIN MS 24171- | Coil data | | | | | | | | | | | Time - (milliseconds maximum) | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Coil | Nominal | | | Max | | Max pick-up voltage | | | Hold voltage 2/ | Drop out voltage 2/ | Operate 3/ | Release 4/ | Bounce | | | |
| | | | | | | | | | | | | | | Main | | Aux | |
| | | Volts 1/ | Freq Hz | Ω Res +15% -10 | Volts | Amp | Normal 2/ | High temp test | Cont current test | | | | | NO | NC | NO | NC |
| D1 | X1,X2 | 28 | dc | 10 | 29 | 4.5 | 7.5 | 11.5 | --- | 3.0 | .5 | 25 | 18 | 5.0 | --- | --- | --- |
| D2 | X1,X2 | 28 | dc | 10 | 29 | 4.5 | 7.5 | 11.5 | --- | 3.0 | .5 | 25 | 18 | 5.0 | --- | --- | --- |

1/ CAUTION: Use of any coil voltage less than nominal coil voltage will compromise the operation of the relay.
2/ Over the temperature range.
3/ With nominal coil voltage.
4/ From nominal coil voltage.

TABLE III. Rated contact load (amperes per pole) case grounded.

| Type of load | Life operating cycles x $10^3$ | 28 V dc | | | | 115 V ac, 1 phase | | | | 115/200 V ac, 3 phase 1/ | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Main | | Aux | | Main | | Aux | | Main | | Aux | |
| | | NO | NC | NO | NC | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz | 400 Hz | 60 Hz |
| Resistive | 50 | 200 | | | | | | | | | | | |
| Inductive | 10 | 100 | | | | | | | | | | | |
| Motor | 50 | 200 | | | | | | | | | | | |
| Lamp | | | | | | | | | | | | | |
| Transfer load | | | | | | | | | | | | | 2/ |
| Mechanical life reduced current | 100 | 50 | | | | | | | | | | | |
| Mixed loads | | | | | | | | | | | | | |

1/ Absence of value indicates relay is not rated for 3-phase application.
2/ Transfer load indicates relay is suitable for transfer between unsynchronized ac power supplies at rating indicated.



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Electrical characteristics:

| | |
|---|---|
| Temperature range | -55°C to +71°C |
| Maximum altitude rating | 50,000 ft |
| Shock G-level | 25 g's |
| Duration | 6-9 ms |
| Max duration contact opening | 2 ms |
| Vibration - sinusoidal (see table IV) | |
| Vibration - Random | N/A |
| High shock | N/A |
| Acceleration | 10 g's |

TABLE IV. Vibration levels.

| 5-10 Hz | 10-55 Hz | 55-250 Hz | 250-500 Hz | 500-1,500 Hz |
|---|---|---|---|---|
| .08 DA | .06 DA | 2 g's | 2 g's | |

Electrical characteristics:

| | |
|---|---|
| Insulation resistance, initial: | 100 megohms. |
| After life or environmental tests: | 50 megohms. |
| Dielectric strength (sea level): | 2-5 seconds. |

| | Initial | | After life tests | |
|---|---|---|---|---|
| | 28 V dc | 115 V ac | 28 V dc | 115 V ac |
| Coil to case | 1,250 V rms | N/A | 1,000 V rms | N/A |
| Aux contacts | 1,250 V rms | N/A | 1,000 V rms | N/A |
| All other points | 1,250 V rms | N/A | 1,000 V rms | N/A |

Dielectric strength (altitude): 1 minute.

| | 28 V dc | 115 V ac |
|---|---|---|
| Coil to case | 500 V rms | N/A |
| Aux contacts | 500 V rms | N/A |
| All other points | 500 V rms | N/A |



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| | |
|---|---|
| Max contact drop initial: | 0.150 volt. |
| After life test: | 0.175 volt. |
| Overload current | (NO) 1,600 amperes. |
| Rupture current | (NO) 2,000 amperes. |
| Duty rating | Intermittent |
| RFI spec | MIL-STD-461 |

(Applicable to coil circuits of ac operated relays)

Qualification by similarity: See MIL-PRF-6106.

Custodian:
Navy - AS
Air Force - 11
DLA - CC

Preparing Activity
DLA - CC

(Project 5945-1206-08)

Review activity:
Navy - EC



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